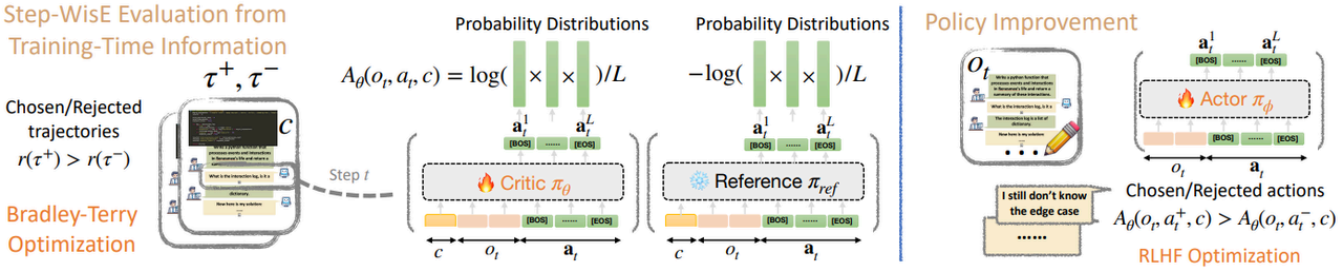
SWEET-RL: Training Multi-Turn LLM Agents on Collaborative Reasoning Tasks

> #DPO #RLHF #Finetuning



- What problems does this framework solve?
  - Large models often face challenges with credit assignment when handling multi-turn dialogues. Traditional reinforcement learning methods (such as PPO or multi-turn DPO) struggle to determine the contribution of intermediate steps—like "confirming user needs"—to the final success, as rewards are typically sparse (only awarded upon completion).
  - Using the single-round RLHF will cause high variance, whereas learning the Value function will prone to overfit when there is limited amount of data.
- What does this framework do?
  - Instead of using a value function, this paper proposes to train an advantage model to predict the advantage.
  - When we train the advantage model (critics), we also attach the information c that is hidden to the actor.
  - Define the advantage as:

    $A_\theta(o_t, a_t, c) = \frac{1}{L}\sum_{l=1}^{L}\left[\log \frac{\pi_\theta(a_t^l|o_t, a_t^{1:l-1}, c)}{\pi_{ref}(a_t^l|o_t, a_t^{1:l-1}, c)}\right]$, the difference in the possibility of the current model θ and the reference model π_{ref}.
  - Instead of using MSE, we take advantage of the training process of DPO: maximize the probability of good trajectories and minimize that of bad trajectories.

    $$\mathcal{J}_{BT} = -\log\sigma\left(\sum_t \beta A_\theta(o_t^+, a_t^+, c) - \sum_t \beta A_\theta(o_t^-, a_t^-, c)\right)$$
    - Intuition: If a trajectory ultimately succeeds, the sum of the advantage values A for each $Action$ composing that trajectory should be higher than that of failing trajectories. This automatically assigns credit through end-to-end gradient backpropagation.
  - Then, with the advantage-based critics model, we evaluate the trajectories generated by the actor, from which we create the +- pair sets to run the normal DPO.

    Here is the step-level (turn-level) comparison.

    With a trained critic, at each dialogue turn t, we have the actor sample multiple distinct candidate actions. The Critic immediately scores these actions, selecting one high-scoring a_t^+ and one low-scoring a_t^-.
- Experiments
  - Benchmark (ColBench): The authors propose a new benchmark, ColBench, comprising two tasks

—“backend programming” and “frontend design”—that simulate the collaborative process between humans and agents in product creation.
    - Significant Results:
        - SWEET-RL achieves approximately 6% higher success rates and win rates compared to existing state-of-the-art multi-turn RL algorithms (e.g., Multi-Turn DPO).
        - Small-scale impact: The Llama-3.1-8B model trained with SWEET-RL matches or even outperforms the much larger Llama-3.1-70B and the closed-source model GPT-4o.
    - Ablation studies: Experiments demonstrate that incorporating “training-time information” (God's-eye view) is crucial for training an accurate Critic; removing this information leads to significant performance degradation.
- Work Cited
    - Paper
    - Chat History
    - NotebookLM